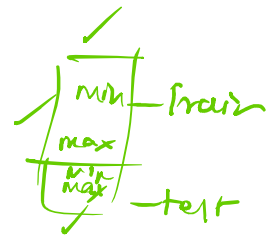


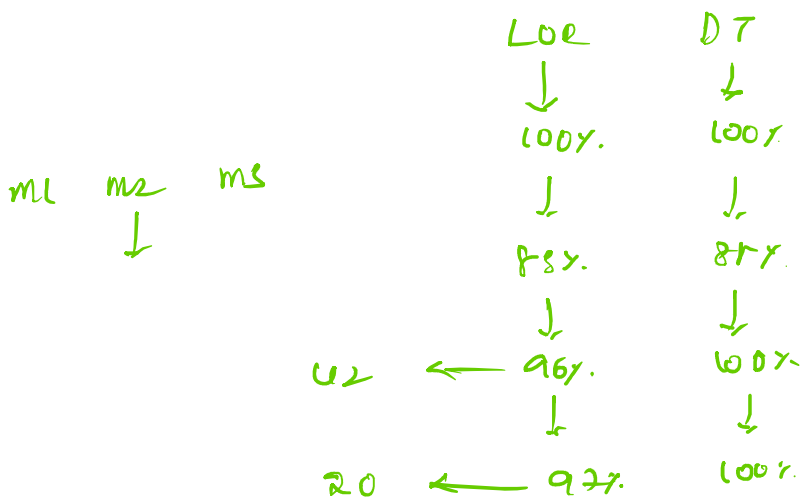
Done with model Building

1. Load dataset
2. Data preprocessing
 - cleaning
 - Analysis
 - Scaling
 - Encoding



data
└── train test

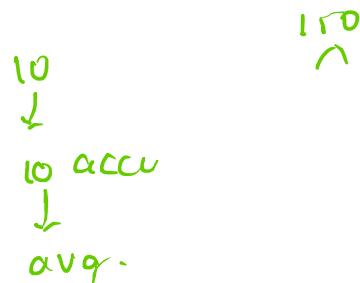
3. Train Test split
4. Fit model
5. predict model
6. Evaluation



Train test sp ($R^2 = 0.92$)

Cross Validation :

— Applying same model multiple time on the same dataset.



150
^

— Underfitting-

low variance ✓

High bias ✓

$$\text{var} \propto \frac{1}{\text{bias}}$$

CV — $\frac{1}{5}$

— Overfitting

low Bias ✓

High variance ✓

✓
— 5

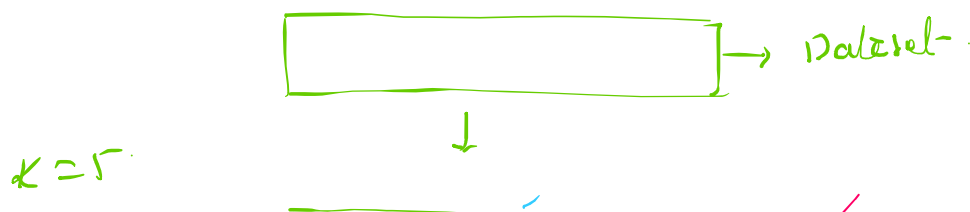
low Bias

low variance -

Types of CV

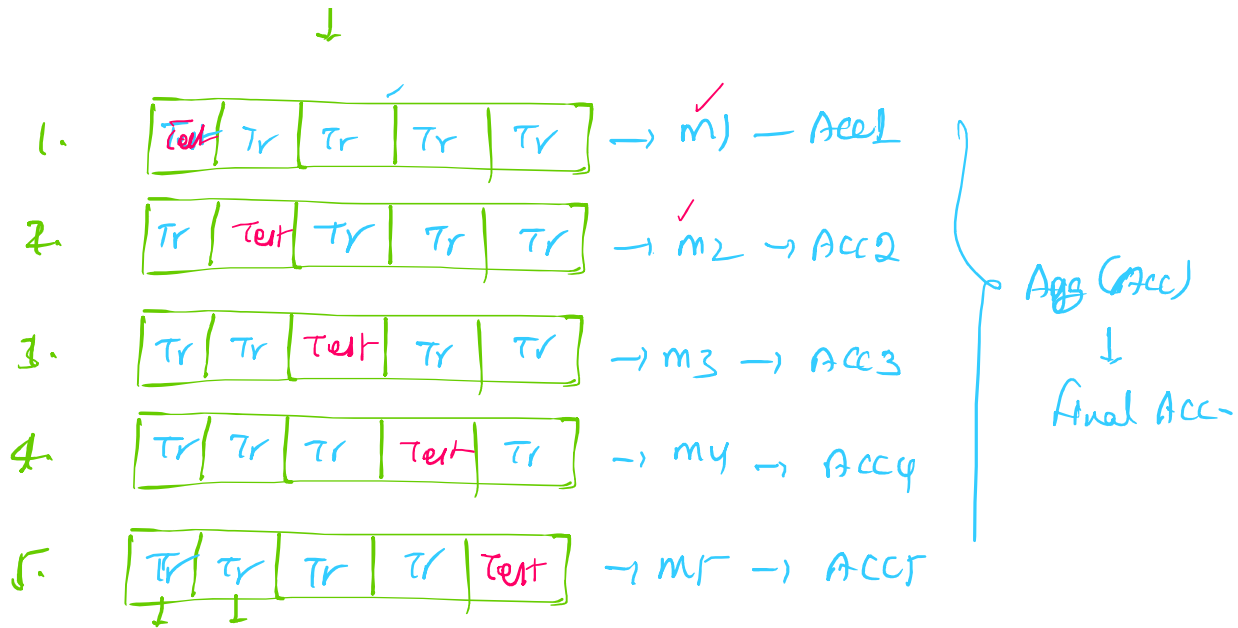
1. K-Fold Cross Validation
2. Stratified K-Fold CV.
3. p-out CV.
4. Repeated KFold CV.

1. K-Fold cross validation:



$k=5$

$$\frac{150}{5} = 30$$



2. Stratified K-Fold:

— Imbalanced data

$$\frac{150}{5} = 30$$

$\swarrow$
 20 yes 10 no



B_i

yes — 100 → 20%

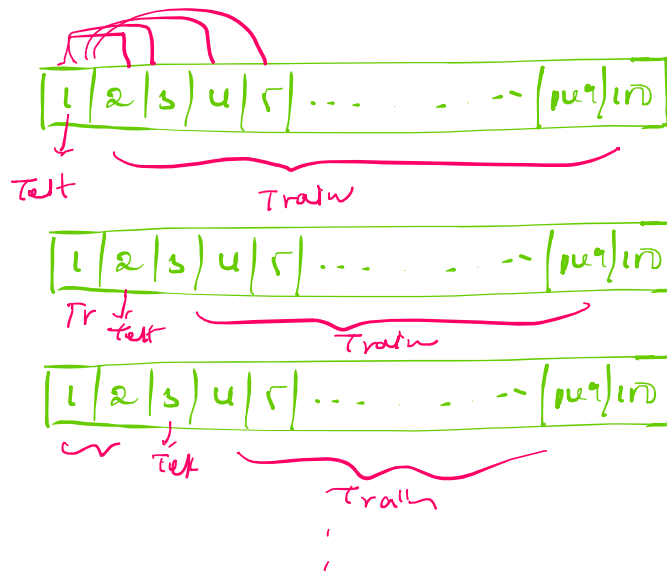
no — 10 — 30%

$$\boxed{30}$$

↓
20.

3. p-out Cross Validation:

$p=1$



$$nCr = \frac{n!}{r!(n-r)!}$$

$$= \frac{150!}{1! \cdot 149!} = 150$$



4. Repeated k-Fold CV:

$$k = 10, R = 5$$

$$10 \times 5 = 50$$